(a) A circular coil of 30 turns and radius 8·0 cm carrying a current of 6 A is suspended vertically in a uniform horizontal magnetic field of 1·0 T. The field lines make an angle of 30° with the plane of the coil. Calculate the magnitude of the external torque that must be applied to prevent the coil from turning. What would happen if the circular coil is replaced by a planar coil of irregular shape that encloses the same area, keeping other parameters unchanged ? *3*

(a) A circular coil of 30 turns and radius 8·0 cm carrying a current of 6 A is suspended vertically in a uniform horizontal magnetic field of 1·0 T. The field lines make an angle of 30° with the plane of the coil. Calculate the magnitude of the external torque that must be applied to prevent the coil from turning. What would happen if the circular coil is replaced by a planar coil of irregular shape that encloses the same area, keeping other parameters unchanged ? *3*

CBSE 2026

$\tau = NIBA \sin\theta$	½
$= 30 \times 6 \times 1 \times 3.14 \times (8 \times 10^{-2})^2 \times \sin 60^\circ$	½
$= 90 \times 64 \times 3.14 \times 1.73 \times 10^{-4}$	½
$= 3.13$ Nm	½
Torque will remain the same.	1

19. A wire of length L is bent round into (i) a square coil having N turns and (ii) a circular coil having N turns. The coil in both cases is free to turn about a vertical axis coinciding with the plane of the coil, in a uniform, horizontal magnetic field and carry the same currents. Find the ratio of the maximum value of the torque acting on the square coil to that on the circular coil. CBSE 2026 **2**

19. A wire of length L is bent round into (i) a square coil having N turns and (ii) a circular coil having N turns. The coil in both cases is free to turn about a vertical axis coinciding with the plane of the coil, in a uniform, horizontal magnetic field and carry the same currents. Find the ratio of the maximum value of the torque acting on the square coil to that on the circular coil. CBSE 2026 **2**

Side of square $a=\frac{L}{4N}$ and radius of circular coil $r=\frac{L}{2\pi N}$	½
$\tau = NIBA$	
$\frac{\tau_1}{\tau_2} = \frac{NIBA_1}{NIBA_2}$	½
$\frac{\tau_1}{\tau_2} = \frac{A_1}{A_2}$	
$\frac{\tau_1}{\tau_2} = \frac{(L/4N)^2}{\pi\left(\frac{L}{2\pi N}\right)^2}$	½
$\frac{\tau_1}{\tau_2} = \frac{\pi}{4}$	½

5. A current carrying loop is placed in a uniform magnetic field. The torque acting on it does ***not*** depend upon the :

(a) magnetic field

(b) current in the loop

(c) area of the loop

(d) shape of the loop

5. A current carrying loop is placed in a uniform magnetic field. The torque acting on it does ***not*** depend upon the :

(a) magnetic field (b) current in the loop

(c) area of the loop (d) shape of the loop

(d) Shape of loop

7. An electron is revolving around a nucleus in a circular orbit of radius r with a constant speed v. The magnetic moment associated with the circulating current is :

(a) $\frac{evr}{4}$

(b) 4 evr

(c) 2 evr

(d) $\frac{evr}{2}$

7. An electron is revolving around a nucleus in a circular orbit of radius r with a constant speed v. The magnetic moment associated with the circulating current is :

(a) $\frac{evr}{4}$

(b) 4 evr

(c) 2 evr

(d) $\frac{evr}{2}$

(d) $\frac{evr}{2}$

18. *Assertion (A) :* A current carrying square loop made of a wire of length L is placed in a magnetic field. It experiences a torque which is greater than the torque on a circular loop made of the same wire carrying the same current in the same magnetic field.

Reason (R) : A square loop occupies more area than a circular loop, both made of wire of the same length.

18. *Assertion (A) :* A current carrying square loop made of a wire of length L is placed in a magnetic field. It experiences a torque which is greater than the torque on a circular loop made of the same wire carrying the same current in the same magnetic field.

Reason (R) : A square loop occupies more area than a circular loop, both made of wire of the same length.

(d) Assertion (A) is false and Reason (R) is also false.

7. The ratio of the magnitude of magnetic moment to that of angular momentum, of an electron revolving around the nucleus in a hydrogen atom is :

(a) Zero

(b) $9 \cdot 27 \times 10^{-24}$ C/kg

(c) $8 \cdot 8 \times 10^{10}$ C/kg

(d) $6 \cdot 6 \times 10^{-12}$ C/kg

7. The ratio of the magnitude of magnetic moment to that of angular momentum, of an electron revolving around the nucleus in a hydrogen atom is :

(a) Zero (b) $9{\cdot}27 \times 10^{-24}$ C/kg

(c) $8{\cdot}8 \times 10^{10}$ C/kg (d) $6{\cdot}6 \times 10^{-12}$ C/kg

(c) 8.8×10^{10} C/kg

18. *Assertion (A) :* When radius of a circular loop carrying a steady current is doubled, its magnetic moment becomes four times.

Reason (R): The magnetic moment of a circular loop carrying a steady current is proportional to the area of the loop.

18. *Assertion (A) :* When radius of a circular loop carrying a steady current is doubled, its magnetic moment becomes four times.

Reason (R): The magnetic moment of a circular loop carrying a steady current is proportional to the area of the loop.

(a) Both Assertion (A) and Reason(R) are true and Reason (R) is the correct explanation of the Assertion (A).

22. A wire of length L is bent round in the form of (i) a square, and then (ii) an equilateral triangle. If current I is passed through each of them, find the ratio of magnetic moment of the square loop to that of the triangle. *2*

For square loop; $L = 4a \Rightarrow a = \frac{L}{4}$ Area of square loop $= \frac{L^2}{16}$ For equilateral triangle; $L = 3a \Rightarrow a = \frac{L}{3}$	½
Area of equilateral triangle $= \frac{\sqrt{3}}{4}a^2$ $= \frac{\sqrt{3}}{4}\left(\frac{L^2}{9}\right) = \frac{\sqrt{3}L^2}{36}$	½
Magnetic moment α Area of the loop.	½
$\Rightarrow \frac{m_s}{m_t} = \frac{L^2}{16} \times \frac{36}{\sqrt{3}L^2} = \frac{3\sqrt{3}}{4}$	½

23. Two wires of equal lengths are shaped in the form of a square loop and a circular loop. Both loops are suspended in a uniform magnetic field. Prove that for the same current, the circular loop will experience larger torque. *2*

Let the length of the wire be l.	
For square loop; $l = 4a \Rightarrow a = \frac{l}{4}$	
Area of square loop $= a^2 = \frac{l^2}{16}$ ---(i)	½
For Circular loop; $l = 2\pi r \Rightarrow r = \frac{l}{2\pi}$	
Area of circular loop(A_c) $= \pi r^2 = \frac{l^2}{4\pi}$ ----(ii)	½
Torque acting on the loop (τ) α A	½
$\because A_c > A_s \quad \therefore \tau_c > \tau_s$	½

19. A wire of length l is in the form of a circular loop A of one turn. This loop is reshaped into loop B of three turns. Find the ratio of the magnetic fields at the centres of loop A and loop B for the same current through them. *2*

$B = \frac{\mu_o NI}{2r}$	½
$l = 2\pi r$; $2\pi r_A = 3(2\pi r_B)$	
$r_B = \frac{r_A}{3}$	½
$\frac{B_A}{B_B} = \frac{\mu_o N_A I}{2r_A} \mathrm{X} \frac{2r_B}{\mu_0 N_B I}$	½
$\frac{B_A}{B_B} = \frac{1}{9}$	½

18. **Assertion (A)** : The deflecting torque acting on a current carrying loop is zero when its plane is perpendicular to the direction of magnetic field. **1**

Reason (R) : The deflecting torque acting on a loop of magnetic moment $\vec{m}$ in a magnetic field $\vec{B}$ is given by the dot product of $\vec{m}$ and $\vec{B}$.

18. **Assertion (A)** : The deflecting torque acting on a current carrying loop is zero when its plane is perpendicular to the direction of magnetic field. **1**

Reason (R) : The deflecting torque acting on a loop of magnetic moment $\vec{m}$ in a magnetic field $\vec{B}$ is given by the dot product of $\vec{m}$ and $\vec{B}$.

(C) Assertion (A) is true and Reason (R) is false.

17. *Assertion (A) :* A planar loop of irregular shape carrying current is subjected to a magnetic field acting perpendicular to the plane of the loop. If the wire is flexible, the loop takes a circular shape.

Reason (R) : The force acting on each point of a current carrying loop, in a magnetic field perpendicular to its plane, is radially outward.

17. *Assertion (A) :* A planar loop of irregular shape carrying current is subjected to a magnetic field acting perpendicular to the plane of the loop. If the wire is flexible, the loop takes a circular shape.

Reason (R) : The force acting on each point of a current carrying loop, in a magnetic field perpendicular to its plane, is radially outward.

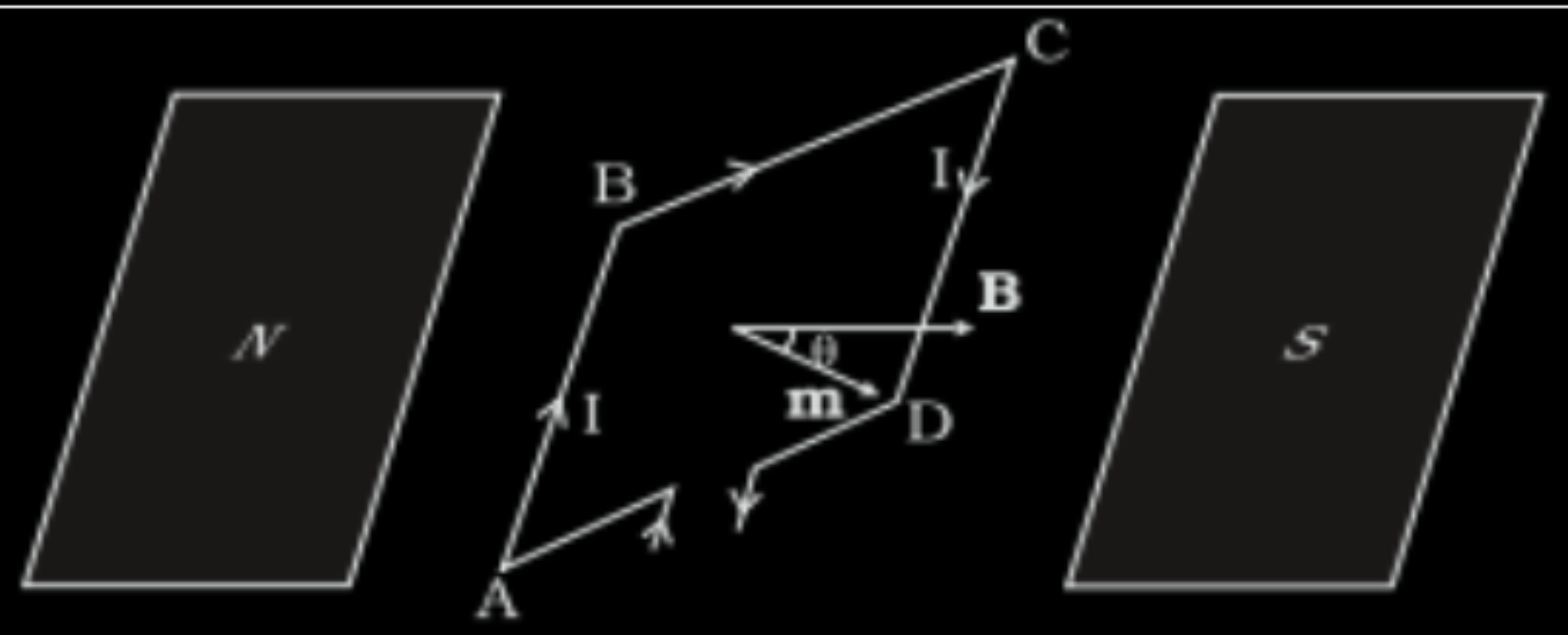
(a)Both Assertion (A) and Reason ® are true, but Reason (R) is the correct explanation of the Assertion (A)

(ii) An electron is revolving around a proton in an orbit of radius r with a speed v. Obtain expression for magnetic moment associated with the electron.

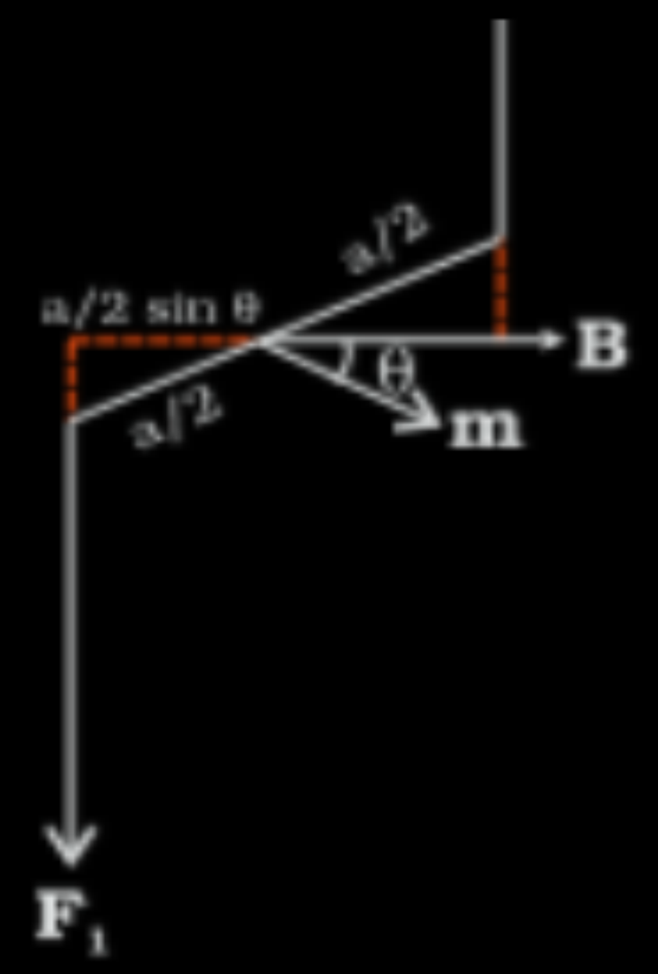
(ii) An electron is revolving around a proton in an orbit of radius r with a speed v. Obtain expression for magnetic moment associated with the electron.

(ii) The equivalent current	
$I = \dfrac{q}{t} = \dfrac{e}{\dfrac{2\pi r}{v}} = \dfrac{ev}{2\pi r}$	½
Mangetic moment of revolving electron	½
$m = IA$	
$= \dfrac{ev}{2\pi r} \times \pi r^2$	½
$= \dfrac{1}{2} evr$	½

23. A rectangular loop of area $\vec{A}$, carrying current I, is placed in a uniform magnetic field $\vec{B}$. With the help of a suitable diagram, derive an expression, in vector form, for the torque acting on the loop. *3*



1



Forces on the arms BC and DA are, equal opposite and collinear. Hence they will cancel each other. ½

The forces on arms AB and CD are $\vec{F_1}$ and $\vec{F_2}$, equal but not collinear. The magnitude of the torque on the loop is

$$\tau = F_1 \frac{a}{2}\sin\theta + F_2 \frac{a}{2}\sin\theta$$ ½

$$= IabB\sin\theta$$

$$= mB\sin\theta \qquad (m = IA)$$ ½

$$\vec{\tau} = \vec{m}\times\vec{B}$$ ½

4. A circular loop of wire, carrying a current 'I' is lying in xy-plane with its centre coinciding with the origin. It is subjected to a uniform magnetic field pointing along + z-axis. The loop will :

(A) move along x-axis (B) move along – y-axis

(C) move along z-axis (D) remain stationary

4. A circular loop of wire, carrying a current 'I' is lying in xy-plane with its centre coinciding with the origin. It is subjected to a uniform magnetic field pointing along + z-axis. The loop will :

(A) move along x-axis (B) move along – y-axis

(C) move along z-axis (D) remain stationary

(D) remain stationary

22. (a) Two long, straight, parallel conductors carry steady currents in opposite directions. Explain the nature of the force of interaction between them. Obtain an expression for the magnitude of the force between the two conductors. Hence define one ampere. *3*

OR

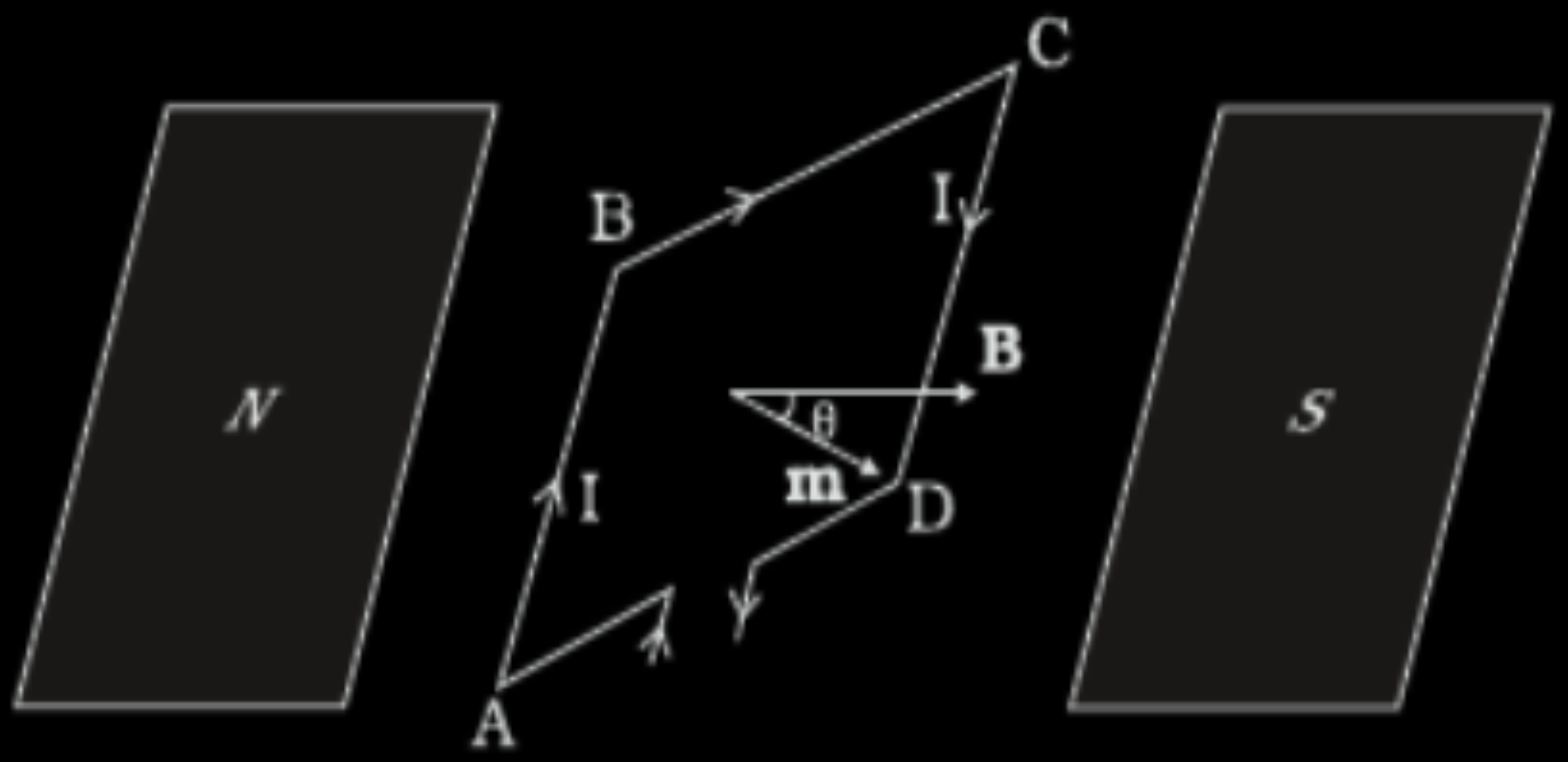
(b) Obtain an expression for the torque $\vec{\tau}$ acting on a current carrying loop in a uniform magnetic field $\vec{B}$. Draw the necessary diagram. *3*

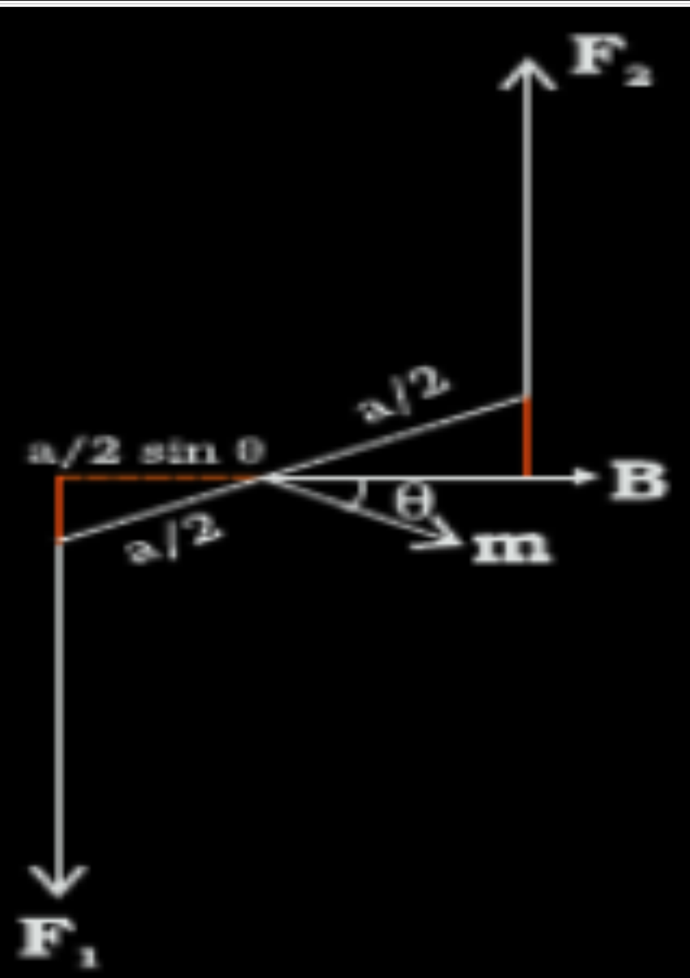
(a)

Explaining nature of force	½
Obtaining expression of force	1½
Defining one ampere	1

(b)

Obtaining expression of torque	2
Drawing diagram	1





1

½

Forces on arm BC and DA are equal and opposite and act along the axis of the coil. Being collinear they cancel each other.

Forces on arms AB and CD are equal and opposite but not collinear. They form a couple. ½

$F_1 = F_2 = IbB$

$$\tau = F_1 \frac{a}{2} \sin\theta + F_2 \frac{a}{2} \sin\theta$$ ½

$\tau = IabB \sin\theta$

$\tau = IAB \sin\theta$ (where A = ab & m = IA)

$\vec{\tau} = \vec{m} \times \vec{B}$ ½

(ii) In a hydrogen atom, the electron moves in an orbit of radius 2 Å making 8×10^{14} revolutions per second. Find the magnetic moment associated with the orbital motion of the electron.

(ii) Finding magnetic moment of electron	2

(ii) In a hydrogen atom, the electron moves in an orbit of radius 2 Å making 8×10^{14} revolutions per second. Find the magnetic moment associated with the orbital motion of the electron.

(ii) Magnetic moment $m = I\,A$	
$I = \frac{e}{T} = ev$	½
$= 1.6\times10^{-19}\times8\times10^{14}$	
$= 1.28\times10^{-4}\,A$	½
$M = 1.28\times10^{-4}\times3.14\times(2\times10^{-10})^2$	½
$= 5.12\pi\times10^{-24}\,Am^2 = 1.6\times10^{-23}\,Am^2$	½

(ii) Finding magnetic moment of electron	2

3. A wire of length 4·4 m is bent round in the shape of a circular loop and carries a current of 1·0 A. The magnetic moment of the loop will be :

(A) $0{\cdot}7 \ \mathrm{Am}^2$

(B) $1{\cdot}54 \ \mathrm{Am}^2$

(C) $2{\cdot}10 \ \mathrm{Am}^2$

(D) $3{\cdot}5 \ \mathrm{Am}^2$

3. A wire of length 4·4 m is bent round in the shape of a circular loop and carries a current of 1·0 A. The magnetic moment of the loop will be :

(A) $0{\cdot}7\ \text{Am}^2$

(B) $1{\cdot}54\ \text{Am}^2$

(C) $2{\cdot}10\ \text{Am}^2$

(D) $3{\cdot}5\ \text{Am}^2$

(B) 1.54Am^2

24. A circular coil with cross-sectional area $0{\cdot}2\ \text{cm}^2$ carries a current of 4 A. It is kept in a uniform magnetic field of magnitude $0{\cdot}5$ T normal to the plane of the coil. Calculate :

(a) the net force on the coil.

(b) the torque on the coil.

(c) the average force on each electron in the coil due to the magnetic field. The free electron density in the material of the coil is $10^{28}\ \text{m}^{-3}$. *3*

	Calculating	
(a)	Net force	1
(b)	Torque	1
(c)	Average force	1

(a)

Net force = zero

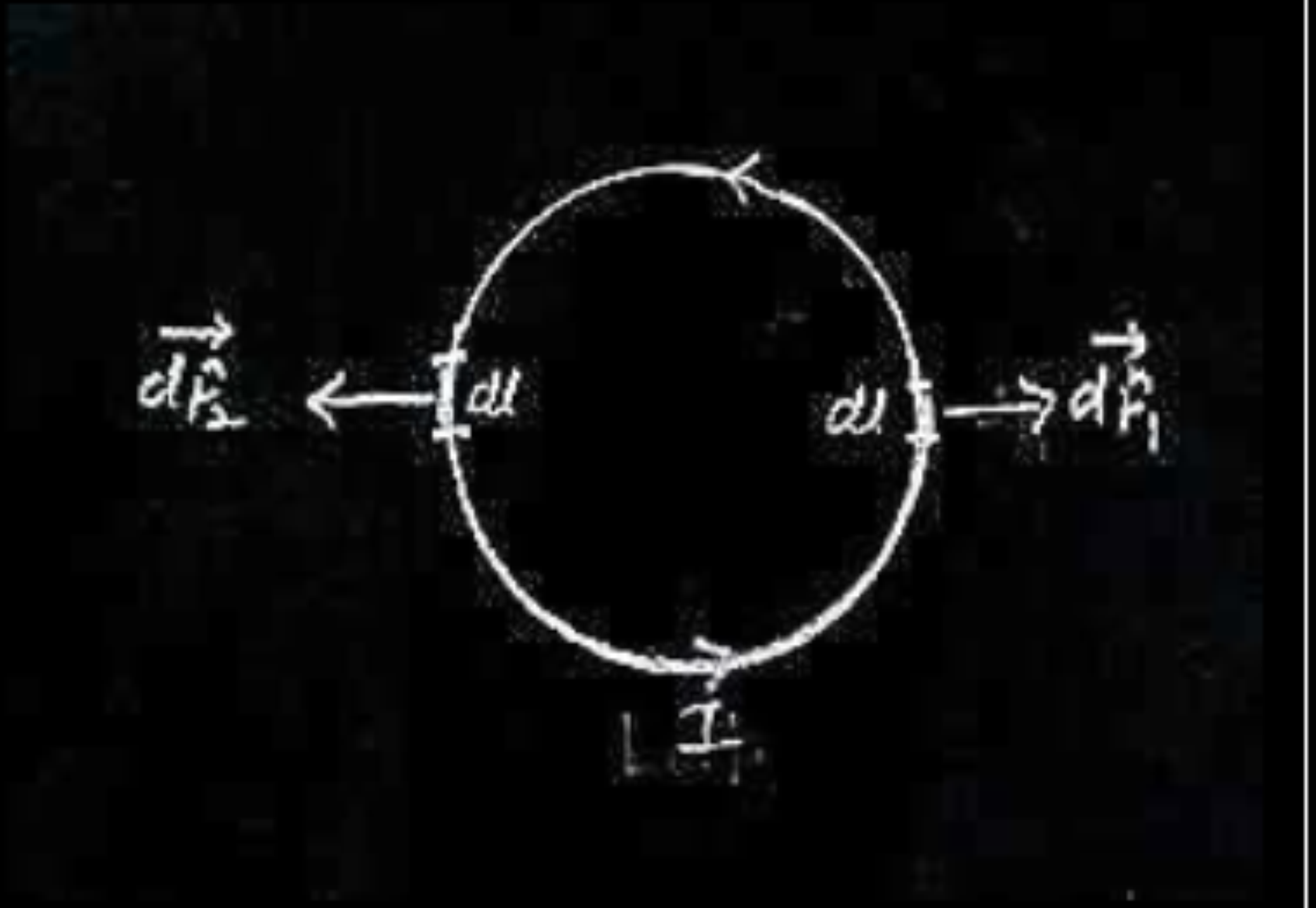
By symmetry, force on each element of the coil is equal and opposite to the force on the diametrically opposite element of the coil. Hence, the net force is zero. 1

Alternatively :

$$d\vec{f}_1 = I\vec{dl} \times \vec{B}$$

$$d\vec{f}_2 = I\vec{dl} \times \vec{B}$$

$$\vec{F}_1 = -\vec{F}_2$$



On coil magnetic force are equal and opposite, so they will cancel out each other.

(b)

Torque on the coil.

$$\vec{\tau} = \vec{m} \times \vec{B}$$

$$\tau = m\,B\sin\theta \qquad \left[\vec{m} \,||\, \vec{B}\right]$$

$$\tau = mB\sin 0^\circ \qquad \theta = 0^\circ$$

$$\tau = 0$$ 1

(c)

$$f_{avg} = e(\vec{v}_d \times \vec{B})$$

$$= e\,v_d\,B\sin 90^\circ$$ ½

$$= e\,v_d\,B$$

$$\therefore\ I = n\,e\,A\,v_d$$

$$v_d = \frac{I}{n\,e\,A}$$

$$f_{avg} = e \times \frac{I}{n\,e\,A} B$$

$$= \frac{IB}{nA}$$

$$= \frac{4 \times 0.5}{10^{28} \times 0.2 \times 10^{-4}}$$ ½

$$= 10^{-23}\ \text{N}$$

4. A loop carrying a current I clockwise is placed in $x - y$ plane, in a uniform magnetic field directed along z-axis. The tendency of the loop will be to :

(A) move along x-axis (B) move along y-axis

(C) shrink (D) expand

4. A loop carrying a current I clockwise is placed in $x - y$ plane, in a uniform magnetic field directed along z-axis. The tendency of the loop will be to :

(A) move along x-axis

(B) move along y-axis

(C) shrink

(D) expand

(C) Shrink

5. A circular loop of radius 'a' carries a current 3I through it. Another concentric circular loop of radius 2a carrying a current I is placed perpendicular to the first loop. The dipole moment of the equivalent magnetic dipole for the system is :

(A) $\pi I a^2$

(B) $5Ia$

(C) $5\pi I a^2$

(D) $7\pi I a^2$

5. A circular loop of radius 'a' carries a current 3I through it. Another concentric circular loop of radius 2a carrying a current I is placed perpendicular to the first loop. The dipole moment of the equivalent magnetic dipole for the system is :

(A) $\pi I a^2$ (B) $5Ia$

(C) $5\pi I a^2$ (D) $7\pi I a^2$

(C) $5\pi I a^2$

16. *Assertion (A) :* The torque acting on a current carrying coil is maximum when it is suspended in a radial magnetic field.

Reason (R) : The torque tends to rotate the coil on its own axis.

16. *Assertion (A) :* The torque acting on a current carrying coil is maximum when it is suspended in a radial magnetic field.

Reason (R) : The torque tends to rotate the coil on its own axis.

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion(A).

(a) An electron is revolving with speed v around the proton in a hydrogen atom, in a circular orbit of radius r. The magnitude of magnetic dipole moment of the electron is : *1*

(A) 4 evr (B) 2 evr

(C) $\frac{1}{2}$ evr (D) $\frac{1}{4}$ evr

(a) An electron is revolving with speed v around the proton in a hydrogen atom, in a circular orbit of radius r. The magnitude of magnetic dipole moment of the electron is : *1*

(A) 4 evr

(B) 2 evr

(C) $\frac{1}{2}$ evr

(D) $\frac{1}{4}$ evr

(C) $\frac{1}{2} evr$

(b) A square loop of side 5.0 cm carries a current of 2.0 A. The magnitude of magnetic dipole moment associated with the loop is : *1*

(A) 1.0×10^{-3} Am^2

(B) 5.0×10^{-3} Am^2

(C) 1.0×10^{-2} Am^2

(D) 5.0×10^{-2} Am^2

(b) A square loop of side 5.0 cm carries a current of 2.0 A. The magnitude of magnetic dipole moment associated with the loop is : *1*

(A) $1.0 \times 10^{-3}\ Am^2$ (B) $5.0 \times 10^{-3}\ Am^2$

(C) $1.0 \times 10^{-2}\ Am^2$ (D) $5.0 \times 10^{-2}\ Am^2$

(B) $5.0 \times 10^{-3}\ Am^2$

(b) (i) Deduce an expression for the torque experienced by a current carrying coil placed in an external uniform magnetic field.

(ii) The maximum torque acting on a current carrying loop of area 0.08 m^2 is 8×10^{-8} N-m. The current in the loop is 1.6 μA. Calculate : *5*

(1) the magnetic field in which the loop is kept, and

(2) the magnetic dipole moment of the loop

(i) Deducing expression for Torque experienced by the coil	3
(ii) Calculating	
(1) The magnetic field	1
(2) The magnetic dipole moment	1

The field exerts no force on the two arms AD and BC of the coil.

The force on arm AB= $F_1 = IbB$ ½

The force on arm CD=F_2=IbB ½

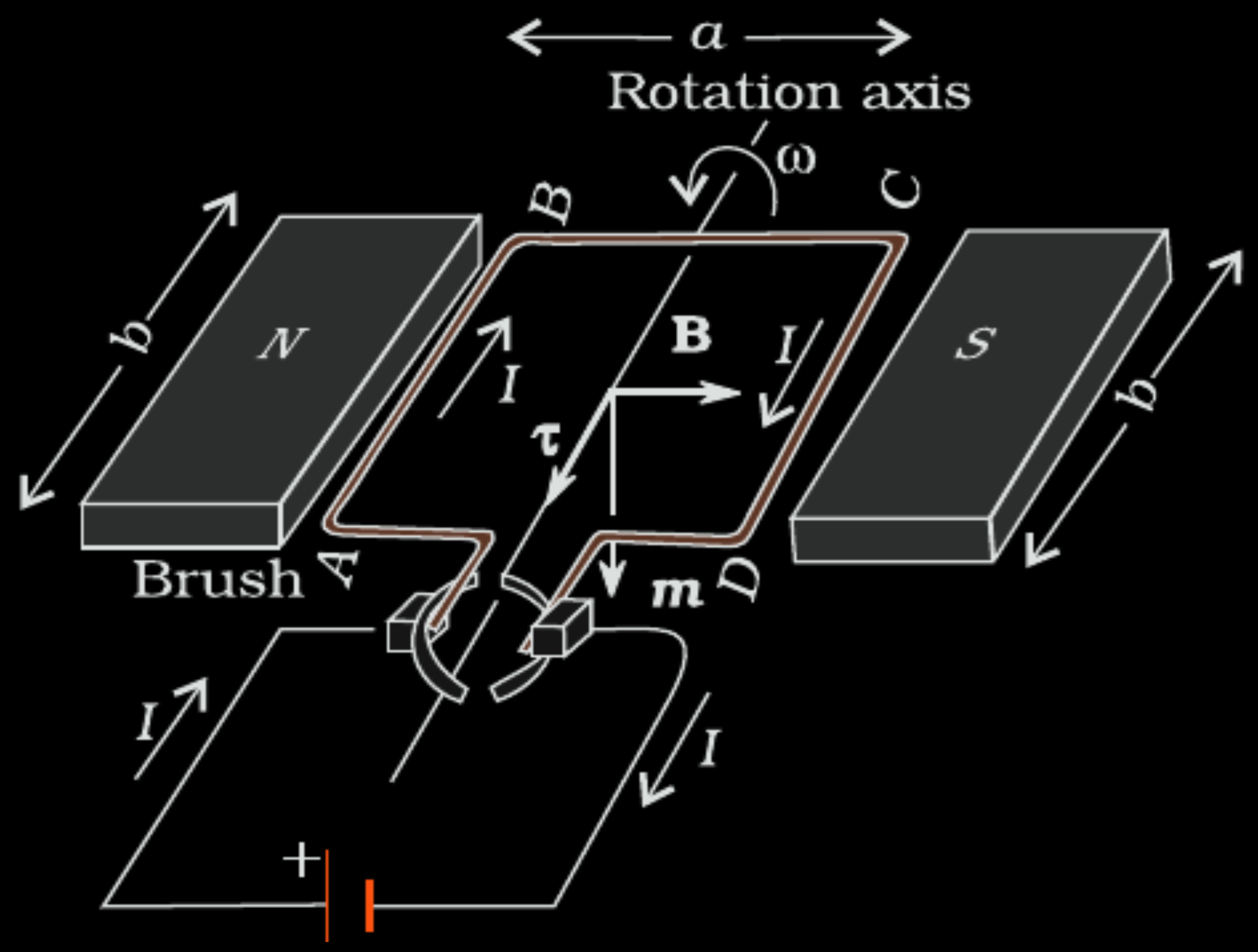
The torque on the coil due to the pair of forces F_1 & F_2 tends to rotate the coil.

Torque acting on the coil,

$$\tau = F_1\frac{a}{2} + F_2\frac{a}{2}$$ ½

$$= IbB\frac{a}{2} + IbB\frac{a}{2}$$ ½

$$\tau = IAB$$ 1



(ii)

(1) Torque $\tau = IAB$ ½

$$B = \frac{\tau}{IA}$$

$$= \frac{8\times10^{-8}}{1.6\times10^{-6}\times0.08}$$

$$B = 0.62T$$ ½

(2) Magnetic dipole moment (m)=IA ½

$m=1.6\times10^{-6}\times0.08$

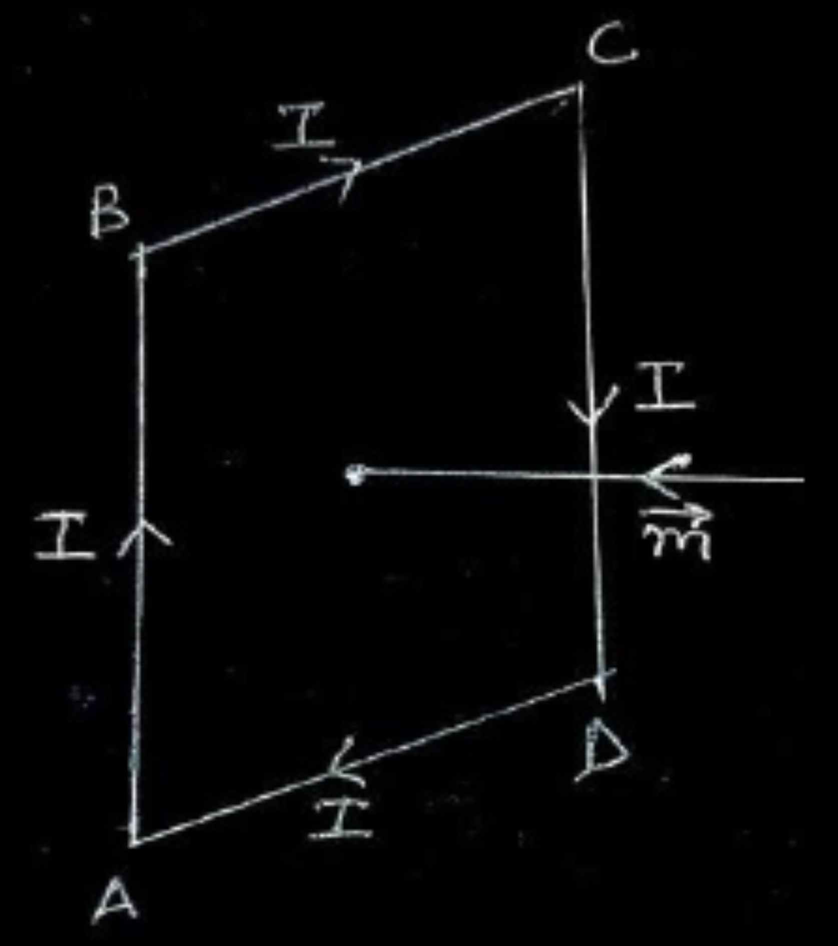
$m=1.28\times10^{-7}\,Am^2$ ½

(b) (i) A rectangular loop of sides l and b carries a current I clockwise. Write the magnetic moment $\vec{m}$ of the loop and show its direction in a diagram.

(ii) The loop is placed in a uniform magnetic field $\vec{B}$ and is free to rotate about an axis which is perpendicular to $\vec{B}$. Prove that the loop experiences no net force, but a torque $\vec{\tau} = \vec{m} \times \vec{B}$. *3*

Writing the expression for magnetic moment & showing its direction	1
Proving no net force	1
Torque $(\vec{\tau}) = \vec{m} \times \vec{B}$	1

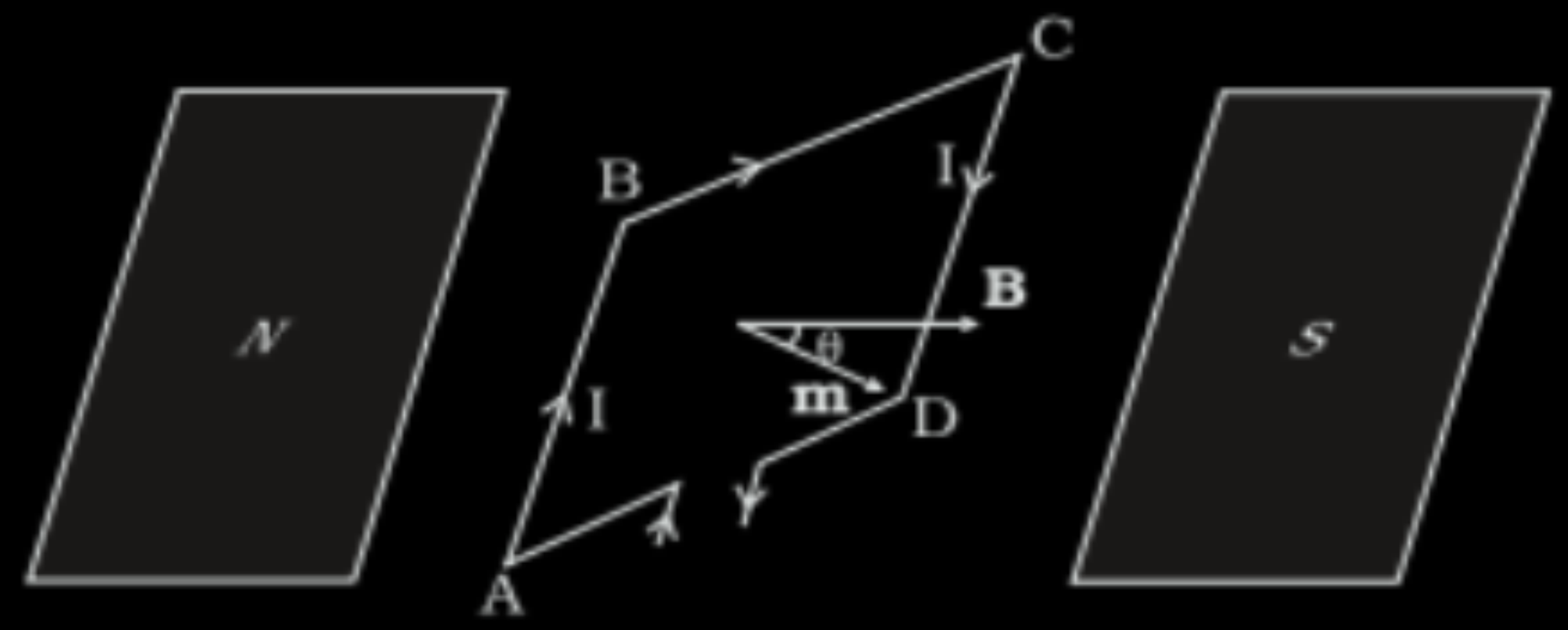
(i) $\vec{m} = I\vec{A}$ ½



½

(ii) $F_1 = F_2 = IbB$ $\quad F_1 =$ Force on AB into the plane

$F_2 =$ Force on CD out of the plane ½



Since forces are equal & opposite so net force = 0 ½

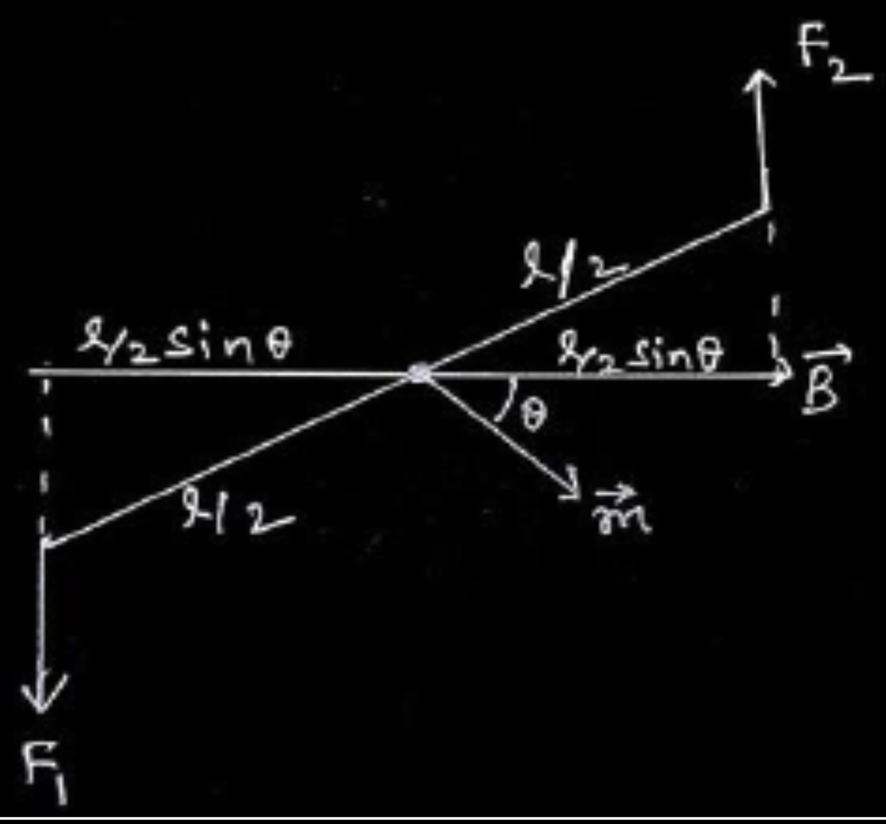
Both forces form a couple, magnitude of torque acting on the coil is

$$\therefore \tau = F_1 \frac{l}{2}\sin\theta + F_2 \frac{l}{2}\sin\theta$$ ½

$= I\, b\, B\, l \sin\theta$

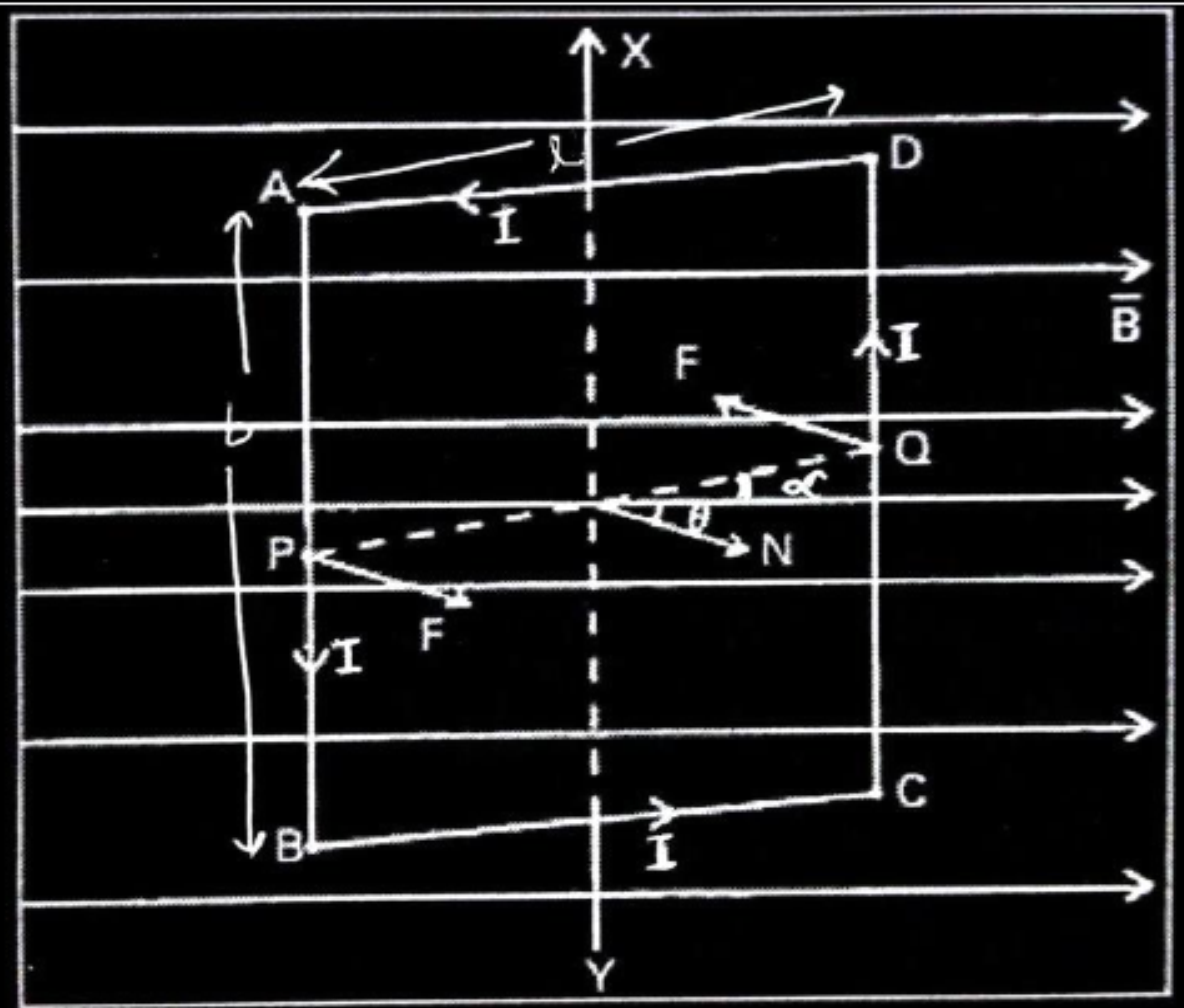
$= I\, A\, B\, \sin\theta$

$= m\, B \sin\theta$



$\vec{\tau} = \vec{m} \times \vec{B}$ ½

Alternatively:



If the plane of the current carrying coil makes an angle ∝ with the magnetic field

$\vec{F}_{DA} = -\vec{F}_{Bc}$ (cancel each other)

Force on the arm DC is into the plane of the paper ½

$|F_{DC}| = IbB$

Force on the arm AB is out of the plane of the paper.

$|F_{AB}| = IbB$

Since forces are equal & opposite so net force = 0 ½

Both of them form a couple and magnitude of torque acting on the coil is

τ=either force ×perpendicular distance between the two forces.

$\tau = IbB \times l \sin\theta$ ½

$= IAB \sin\theta$

$\vec{\tau} = I\vec{A} \times \vec{B}$ ½

$\vec{\tau} = \vec{m} \times \vec{B}$

32. (a) (A) When a current passes through a conductor it produces a magnetic field.

(i) State the right-hand thumb rule that gives the direction of the magnetic field.

(ii) How does the magnetic field vary with the distance from the conductor ?

(iii) Name the factors (other than the distance) which affect the magnitude of the magnetic field.

(B) Explain how a magnetic dipole in a uniform magnetic field attains (i) stable equilibrium (ii) unstable equilibrium. **5**

(A) (i) Stating the right-hand thumb rule	1
(ii) Variation of magnetic field with distance	1
(iii) Naming the factors affecting magnitude of magnetic field	1
(B) Explanation of magnetic dipole to attain	
(i) Stable equilibrium	1
(ii) Unstable equilibrium	1

(a) (i) Right hand thumb rule When we hold a conductor in right hand in such a way that a thumb stretches in the direction of flow of current than the curled fingers shows the direction of magnetic field.	1
(ii) Magnetic field due to straight conductor at a distance r is given by Since $B \propto \frac{1}{r}$ As r increases , B decreases	1
(iii) B depends on current ,nature of medium and angle θ between current element and displacement r.(any two factors)	½ +½
(B) Explanation (i) Stable equilibrium When the magnetic dipole aligned along the direction of external magnetic field.	1
(ii) Unstable equilibrium When the magnetic dipole align in opposite direction of external magnetic field .	1

(B) Derive an expression for torque acting as a rectangular loop carrying a steady current when placed in a uniform magnetic field.

(B) Derivation of torque The magnetic field exerts no force on two arms of length 'a' parallel to magnetic field.	½
The field exerts a force on two arm of length b perpendicular to magnetic field is, $F_1 = F_2 = IbB$	½
F_1 is directed into the plane of the loop and F_2 is directed out of the plane of paper. Thus the net force on the loop is zero .	½
There is a torque on loop due to pair of force $\vec{F_1}$ and $\vec{F_2}$. The torque on the loop tends to rotate anticlockwise. The magnitude of torque - $\tau = F_1 \frac{a}{2} + F_2 \frac{a}{2}$	½
$= IbB\frac{a}{2} + IbB\frac{a}{2} = I(ab)B$	½
$\tau = IAB$ Where A=ab is the area of rectangle.	

7. A current of $\left(\frac{10}{\pi}\right)$A is maintained in a circular loop of radius 14 cm. The value of dipole moment associated with the loop is :

(A) $0{\cdot}019\ \mathrm{Am}^2$

(B) $0{\cdot}14\ \mathrm{Am}^2$

(C) $0{\cdot}196\ \mathrm{Am}^2$

(D) $0{\cdot}615\ \mathrm{Am}^2$

7. A current of $\left(\frac{10}{\pi}\right)$A is maintained in a circular loop of radius 14 cm. The value of dipole moment associated with the loop is :

(A) $0{\cdot}019\ \text{Am}^2$

(B) $0{\cdot}14\ \text{Am}^2$

(C) $0{\cdot}196\ \text{Am}^2$

(D) $0{\cdot}615\ \text{Am}^2$

(C) $0.196\ \text{Am}^2$

26. An electron of mass m and charge $-e$ is revolving anticlockwise around the nucleus of an atom.

(a) Obtain the expression for the magnetic dipole moment (μ) of the atom.

(b) If $\vec{L}$ is the angular momentum of electron, show that

$$\vec{\mu} = -\left(\frac{e}{2m}\right)\vec{L}.$$ 3

a)	Obtaining expression for magnetic dipole moment	1½
b)	To Show $\vec{\mu} = -\left(\frac{e}{2m}\right)\vec{L}$	1½

a) $\mu = IA$	½
$= \frac{e}{T} \times A$	
$= \frac{e}{\frac{2\pi r}{v}} \times \pi r^2$	½
$= \frac{1}{2} evr$	½
b) $L = mvr$	½
$\mu = \frac{evr \times m}{2 \times m}$	
$= \left(\frac{e}{2m}\right) L$	½
Direction of $\vec{\mu}$ is opposite to that of $\vec{L}$	
$\vec{\mu} = -\left(\frac{e}{2m}\right) \vec{L}$	½

(b) (i) A current carrying loop can be considered as a magnetic dipole placed along its axis. Explain.

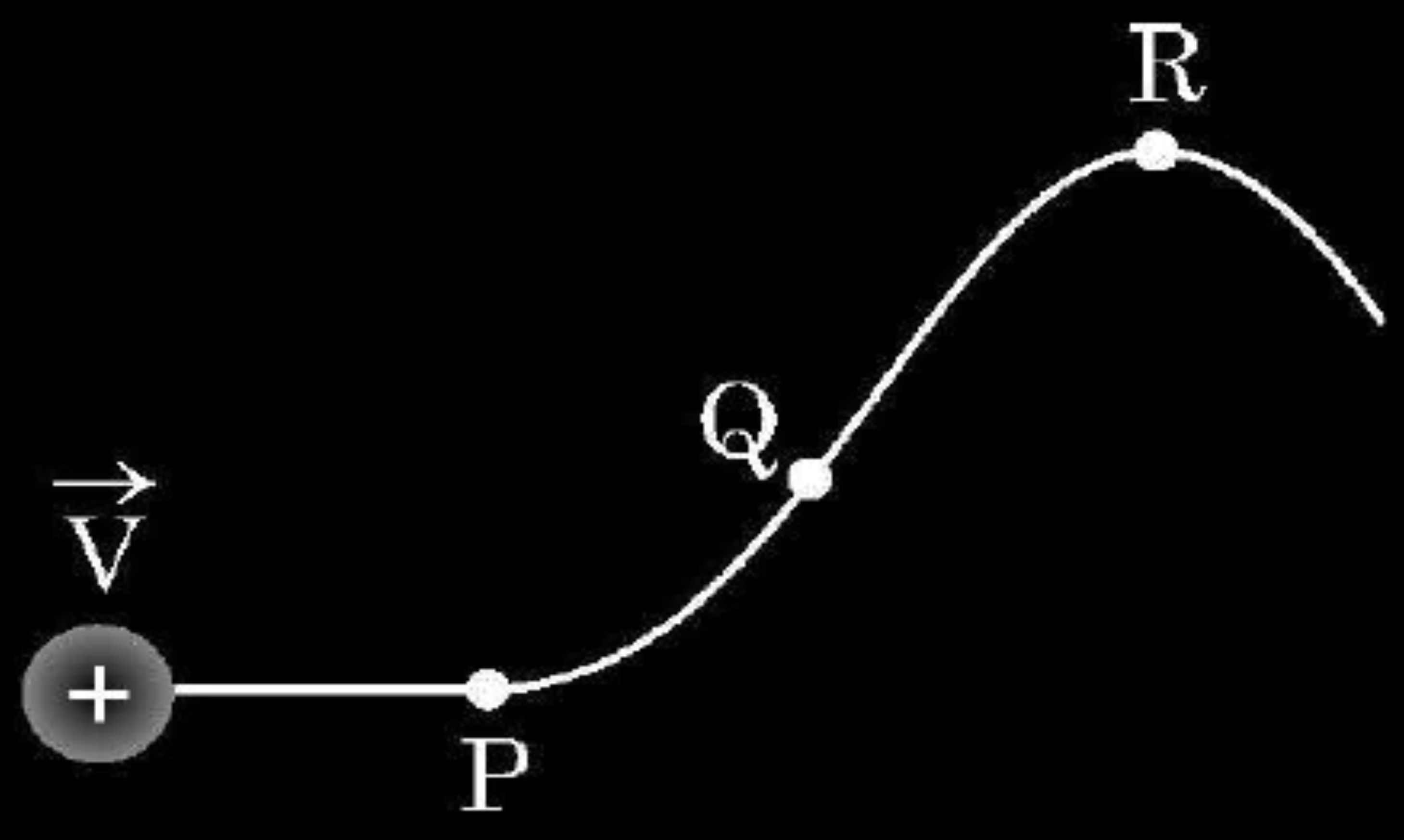
(ii) Obtain the relation for magnetic dipole moment $\vec{M}$ of current carrying coil. Give the direction of $\vec{M}$.

(iii) A current carrying coil is placed in an external uniform magnetic field. The coil is free to turn in the magnetic field. What is the net force acting on the coil ? Obtain the orientation of the coil in stable equilibrium. Show that in this orientation the flux of the total field (field produced by the loop + external field) through the coil is maximum.

(i) Explanation	1
(ii) Obtaining relation for $\vec{M}$, and direction of $\vec{M}$.	1+1
(iii) Net force on coil	1
Obtaining orientation	½
Showing flux is maximum	½

(i) The two faces of a current carrying loop behave like two poles of a magnet therefore can be considered as a magnetic dipole placed along its axis.	1
(ii) Magnetic moment (M) α Current (I) α Area (A) $\therefore\ \vec{M} = I\vec{A}$	1
Direction is same as the area vector.	1
Alternatively: - Magnetic moment is perpendicular to the plane of the coil.	
(iii) Net force acting on the coil is zero.	1
The potential energy (U_B) of a current carrying loop in an external magnetic field $= -\vec{M}.\vec{B}$	
For the coil to be in stable equilibrium U_B should be minimum so $\theta = 0^o$.	½
Therefore, magnetic flux (ϕ) due to the total field $= (B_{coil} + B_{ext})A$, which is its maximum value.	½
Alternatively: - Orientation of stable equilibrium is one where the area vector A of the loop is in the direction of external magnetic field. In this orientation, the magnetic field produced by the loop is in the same direction as external field, both normal to the plane of the loop, thus giving rise to maximum flux of the total field.	

33. (a) (i) A proton moving with velocity $\vec{V}$ in a non-uniform magnetic field traces a path as shown in the figure. **5**

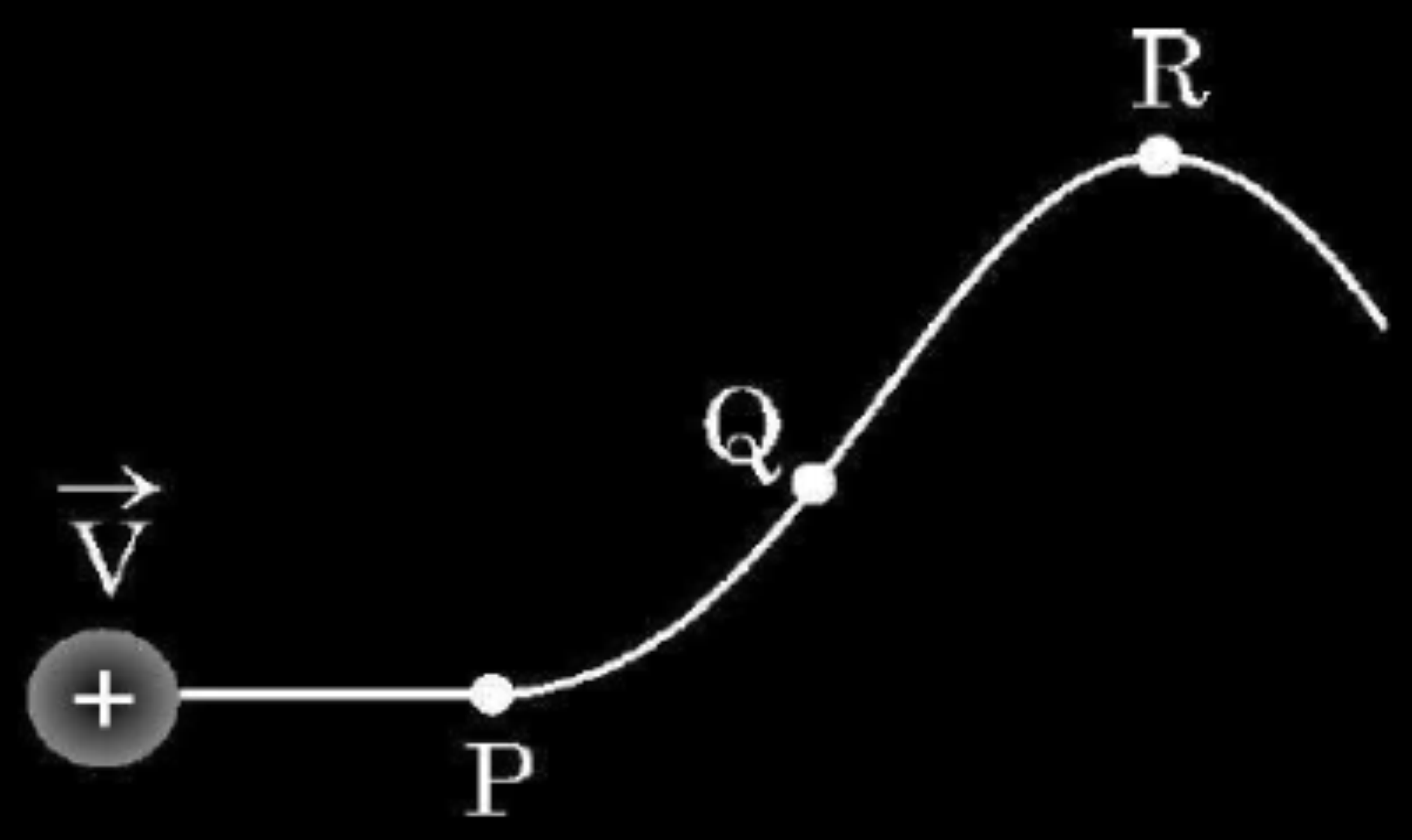


The path followed by the proton is always in the plane of the paper. What is the direction of the magnetic field in the region near points P, Q and R ? What can you say about relative magnitude of magnetic fields at these points ?

(ii) A current carrying circular loop of area A produces a magnetic field B at its centre. Show that the magnetic moment of the loop is $\dfrac{2\,BA}{\mu_0}\sqrt{\dfrac{A}{\pi}}$.

i) Finding the direction of magnetic field near points P,Q and R	½ + ½ + ½
Conclusion about the relative magnitude of magnetic field.	½ + ½ + ½
ii) Showing the given expression of magnetic moment.	2

i) Near point P	
Magnetic field is acting into the plane of the paper as Force is acting upwards.	½
Near point Q	
Magnetic field is into the plane of paper as force is acting upwards.	½
Near point R	
Magnetic field is acting out of the plane of the paper as $\vec{F}$ is acting downwards.	½
Relative Magnitude of the Magnetic field.	
As B $\alpha \frac{1}{r}$	
Therefore,	
Near point P, magnitude of B is small.	½
Near point Q, B is relatively smaller than point P.	½
Near point R, B is relatively larger than point P.	½
$(B_Q < B_P < B_R)$	
ii) Let r be the radius of the circular coil and I is the current in the coil then	
$B = \frac{\mu_0 I}{2r}$ or $I = \frac{2Br}{\mu_0}$	½
$A = \pi r^2$ $\quad r = \sqrt{\frac{A}{\pi}}$	½
$M = IA$	½
$= \frac{2Br}{\mu_0} A$	
$= \frac{2BA}{\mu_0}\sqrt{\frac{A}{\pi}}$	½

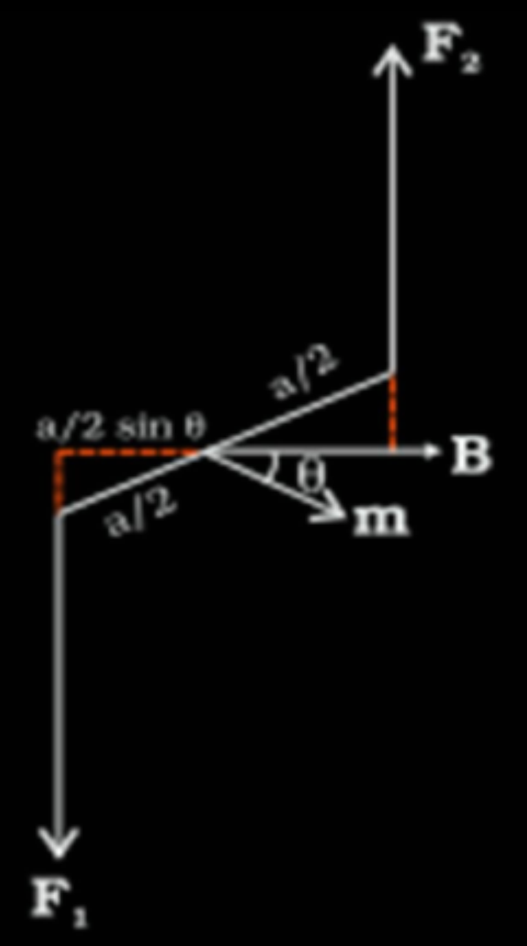


(b) (i) Derive an expression for the torque acting on a rectangular current loop suspended in a uniform magnetic field.

(ii) A charged particle is moving in a circular path with velocity $\vec{V}$ in a uniform magnetic field $\vec{B}$. It is made to pass through a sheet of lead and as a consequence, it looses one half of its kinetic energy without change in its direction. How will (1) the radius of its path (2) its time period of revolution change ?

i) Deriving the expression for the torque.	3
ii) 1) Finding the change in radius.	1
2) Finding the change in time period of Revolution.	1

i)



½

$\vec{F}_1$ *and* $\vec{F}_2$ are the forces acting on two arms of the rectangular coil having sides a and b.

$\left|\vec{F}_1\right| = \left|\vec{F}_2\right| = I\,b\,B$ (b = length of the arm) 1

Forces constitute a couple. The magnitude of Torque on the loop is –

$\tau = F_1 \dfrac{a}{2}\sin\theta + F_2 \dfrac{a}{2}\sin\theta$ ½

$= I\,a\;b\,B\sin\theta$ ½

$= IA\,B\sin\theta$

$\vec{\tau} = I\vec{A}\times\vec{B}$ ½

ii) 1) $r = \dfrac{mv}{qB} = \dfrac{\sqrt{2mK}}{qB}$ ½

$r \;\alpha\; \sqrt{K}$

$\dfrac{r'}{r} = \dfrac{\sqrt{K/2}}{\sqrt{K}} = \dfrac{1}{\sqrt{2}}$

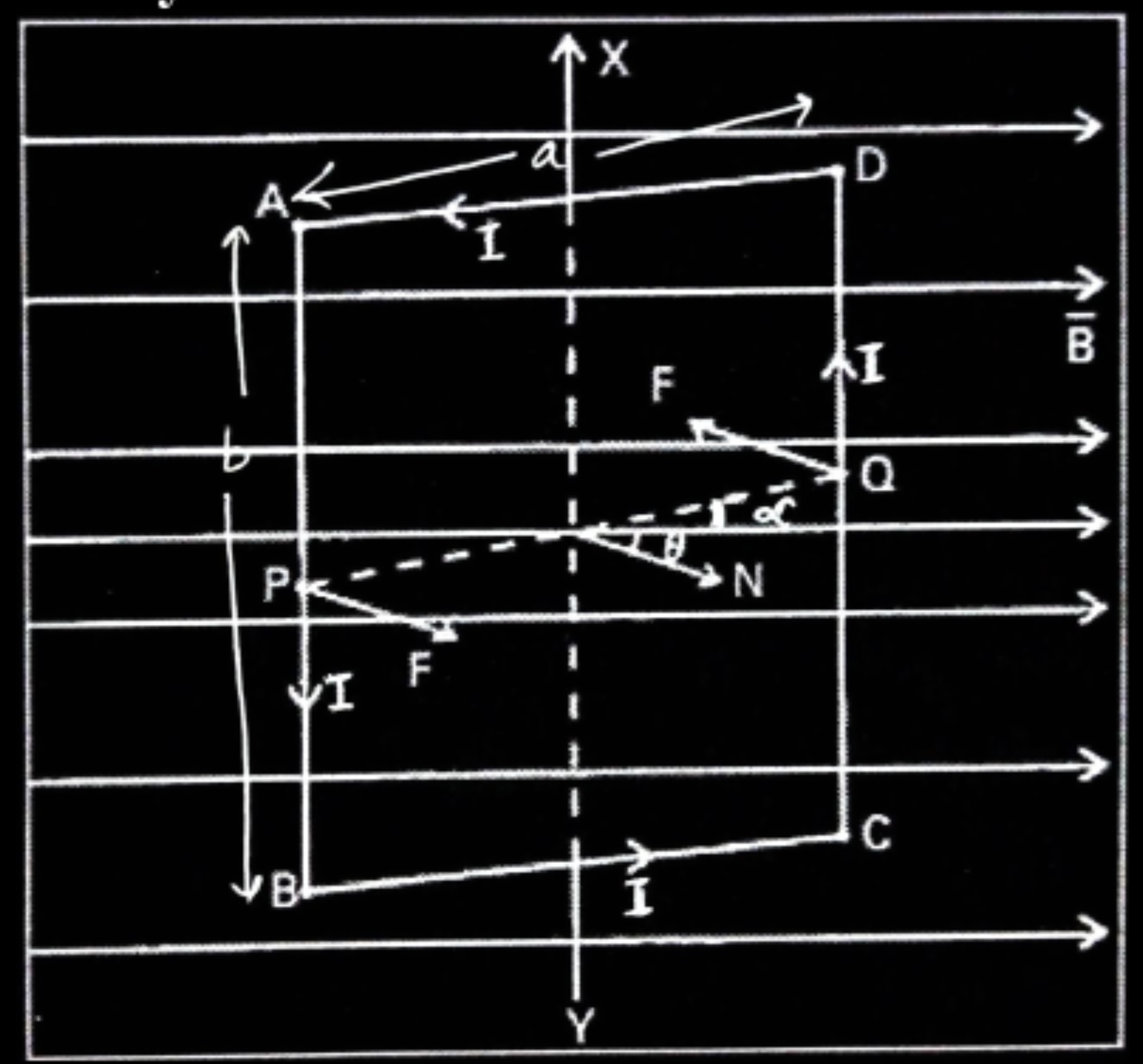
$r' = \dfrac{r}{\sqrt{2}}$ ½

2) $T = \dfrac{2\pi m}{qB}$ ½

Time period does not depend on Kinetic Energy

∴ Time period will not change. ½

Alternatively



½

If the plane of the current carrying coil marks an angle ∝ with the magnetic field

$\vec{F}_{DA} = -\vec{F}_{Bc}$ (cancel each other) . ½

Force on the arm DC is into the plane of the paper

$\left|F_{DC}\right| = IbB$. ½

Force on the arm AB is out of the plane of the paper.

$\left|F_{AB}\right| = IbB$ ½

Both of them form a couple and Torque acting on the coil is

τ=either force ×perpendicular distance between the two forces.

$\tau = IbB \times a\cos\alpha$

$= IabB\cos\alpha$

$\tau = IAB\cos\alpha$ ½

Let $\hat{n}$ =outward drawn normal to the plane of the coil.

$\theta + \alpha = 90^0$

$\alpha = 90^0 - \theta$

$\tau = IAB\cos(90-\theta)$

$= IAB\sin\theta$ ½

$\vec{\tau} = I\vec{A}\times\vec{B}$

23. (a) Define magnetic moment of a current-carrying coil. Write its SI unit.

(b) A coil of 60 turns and area 1.5×10^{-3} m^3 carrying 2A current lies in a vertical plane. It experiences a torque of 0.12 Nm when placed in a uniform horizontal magnetic field. The torque acting on the coil changes to 0.05 Nm after the coil is rotated about its diameter by 90°, in the magnetic field. Find the magnitude of the magnetic field. **3**

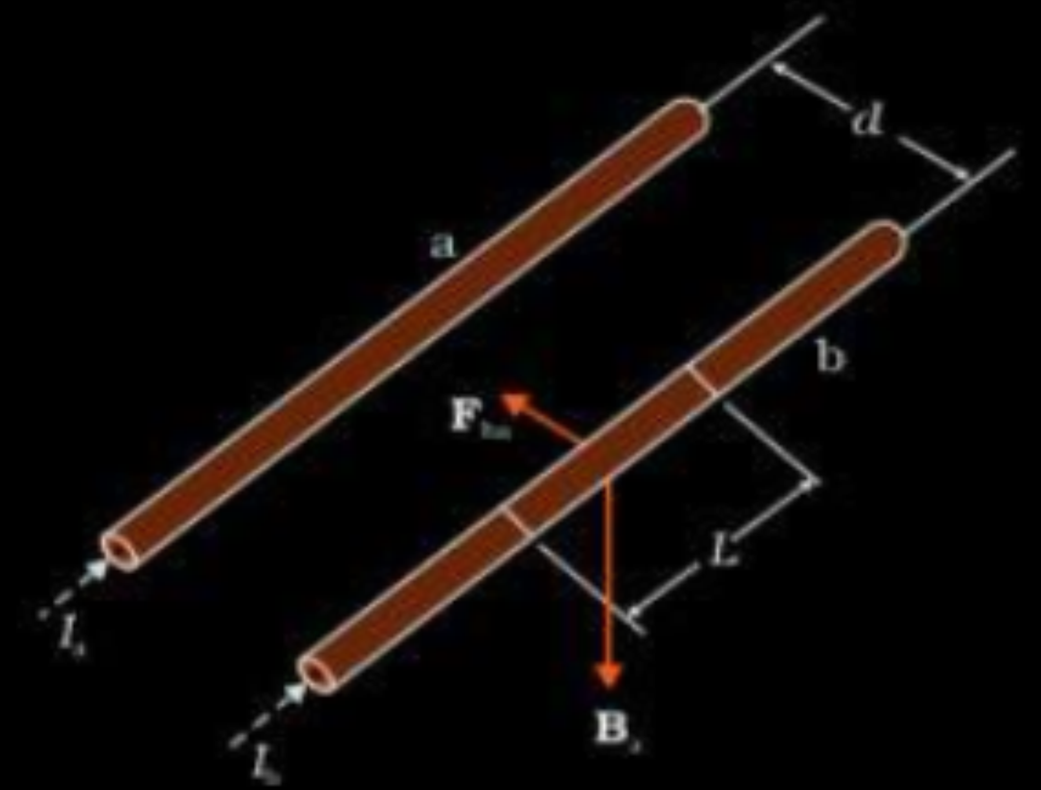
a) Defining magnetic moment	1
SI unit of magnetic moment	½
b) Finding the magnitude of magnetic field	1½

a) Magnetic moment of a current carrying coil is defined as the product of current flowing through the coil and Area of the coil. **Alternatively** $\vec{M} = I\vec{A}$ S.I. unit is Am^2	1 ½
b) $\tau = NIABsin\theta$ $\tau_1 = 0.12 = 60 \times 2 \times 1.5 \times 10^{-3} \times B\sin\theta$ $B\sin\theta = \frac{2}{3}$ $\tau_2 = 0.05 = 60 \times 2 \times 1.5 \times 10^{-3} \times B\cos\theta$ $B\cos\theta = \frac{5}{18}$ $B = \sqrt{B^2\sin^2\theta + B^2\cos^2\theta}$ $= \sqrt{\left(\frac{2}{3}\right)^2 + \left(\frac{5}{18}\right)^2} = \frac{13}{18}T$	 ½ ½ ½

(a) Derive the expression for the force acting between two long parallel current carrying conductors. Hence, define 1 A current.

(b) A steady current of 2A flows through a circular coil having 5 turns of radius 7 cm. The coil lies in X-Y plane with its centre at the origin. Find the magnitude and direction of the magnetic dipole moment of the coil. **5**

(a)



½

Magnetic field due to the current I_a flowing in conductor 'a' at any point on conductor 'b'

$$B_a = \frac{\mu_0 I_a}{2\pi d}$$ ½

(Acting perpendicular downward)

Therefore force on conductor 'b' due to field B_a

$$\vec{F} = I_b (\vec{l_b} \times \vec{B_a})$$ ½

$$|\vec{F_{ba}}| = I_b l_b \times \frac{\mu_0 I_a}{2\pi d}$$

$$= \frac{\mu_0 I_a I_b l_b}{2\pi d}$$

$$\frac{|\vec{F_{ba}}|}{l_b} = \frac{\mu_0 I_a I_b}{2\pi d}$$ ½

Definition of 1 A :

Two straight infinitely long parallel conductors are said to carry 1 A current each when they interact each other with a force of $2 \times 10^{-7}\, Nm^{-1}$, when kept 1m apart in vacuum 1

(b) $M = NI\pi a^2$

$= 5 \times 2 \times \frac{22}{7} \times 49 \times 10^{-4}$ ½

$= 154 \times 10^{-3}$ ½

$= 0.154\ Am^2$ ½

$\vec{M}$ *will be perpendicular to* $x - y$ *plane or parallel to* Z *axis* ½

5. The magnetic dipole moment of a current carrying coil does ***not*** depend upon *1*

(A) number of turns of the coil.

(B) cross-sectional area of the coil.

(C) current flowing in the coil.

(D) material of the turns of the coil.

5. The magnetic dipole moment of a current carrying coil does ***not*** depend upon *1*

(A) number of turns of the coil.

(B) cross-sectional area of the coil.

(C) current flowing in the coil.

(D) material of the turns of the coil.

(D)

material of the turns of the coil

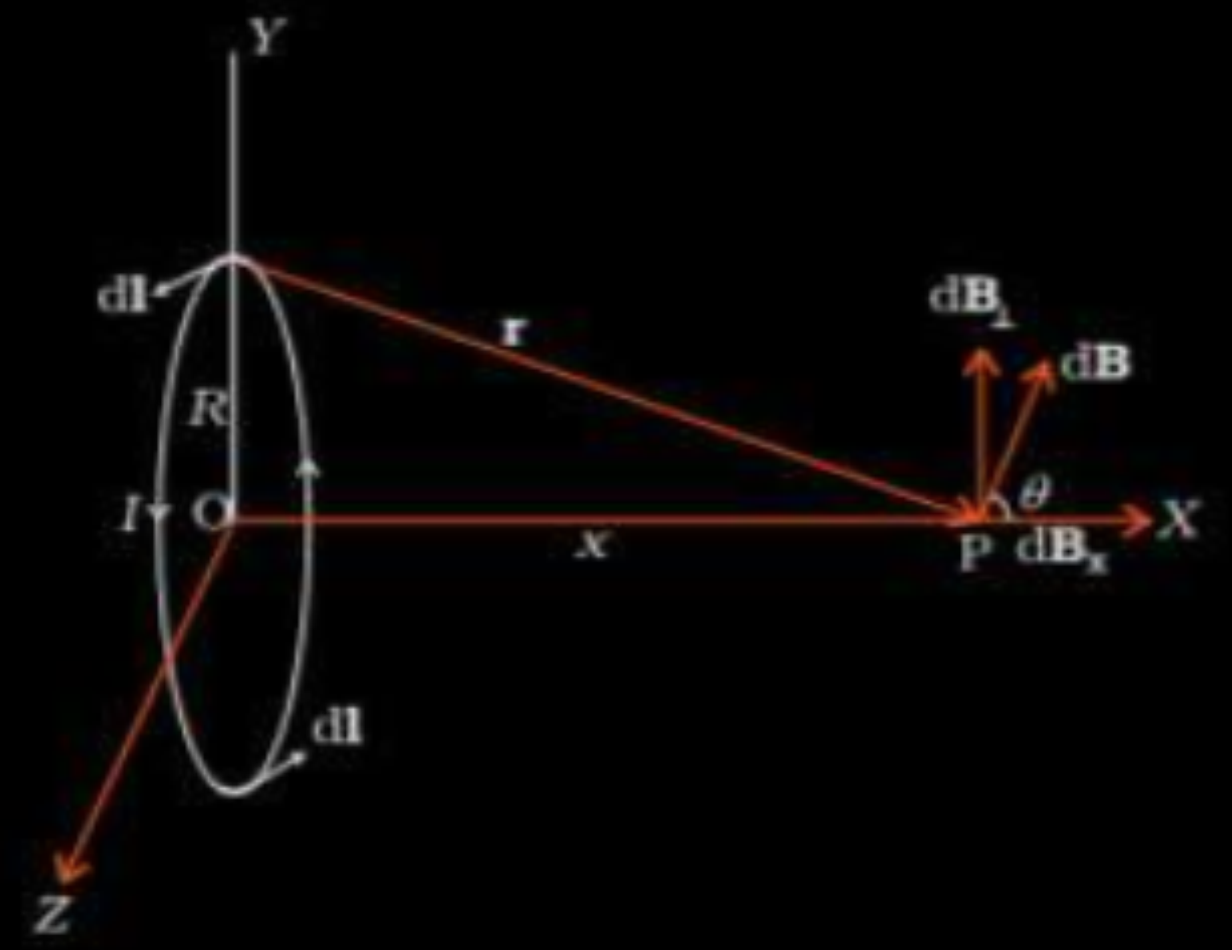
29. (a) Write an expression of magnetic moment associated with a current (I) carrying circular coil of radius r having N turns.

(b) Consider the above mentioned coil placed in YZ plane with its centre at the origin. Derive expression for the value of magnetic field due to it at point (x, 0, 0). *3*

a) Writing expression for magnetic moment	½ mark
b) Figure	½ mark
Magnetic field and calculation	2 mark

(a) magnetic moment = M= NIA ½

$$M = NI\pi r^2$$



½

According to Biot-sevart law

$$\overrightarrow{dB} = \frac{\mu_0 I}{4\pi}\frac{|\overrightarrow{dl} \times \vec{r}|}{r^3}$$ ½

$$dB = \frac{\mu_0 I}{4\pi}\frac{dl}{r^2}$$

$dB_\perp$ components due to diametrically opposite components cancel out. Only dB_x components refrain

$$dB_x = \frac{\mu_0 I dl}{4\pi r^2}.cos\theta$$ ½

$$B = \int dB_x$$

$$B = \frac{\mu_0 I R^2}{2(R^2 + x^2)^{3/2}}\ (along\ x\ axis)$$ 1

(a) Derive the expression for the torque acting on a current carrying loop placed in a magnetic field.

(b) Explain the significance of a radial magnetic field when a current carrying coil is kept in it.

(a)	Derivation of the expression for torque	2
(b)	Significance of radial magnetic field	1

Force on arm AB = F_1 = IbB (directed into the plane of the loop) ½
Force on arm CD = F_2= IbB (directed out of the plane of the loop)

Therefore the magnitude of the torque on the loop due to these pair of forces

$$\tau = F_1 \frac{a}{2} + F_2 \frac{a}{2}$$ ½

= I (ab) B

= IAB = mB ½

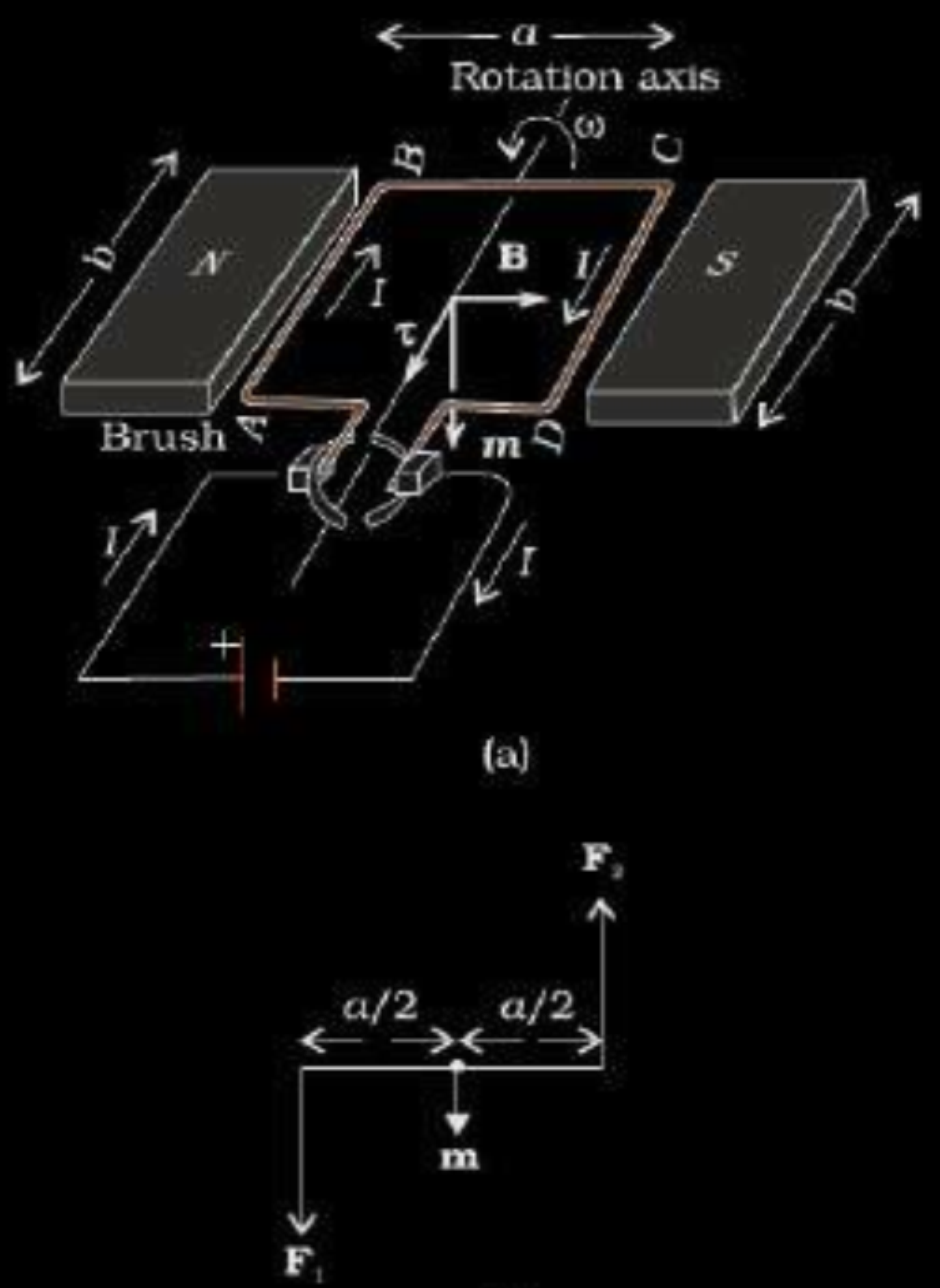
(A = ab = area of the loop)

<u>Alternatively</u>

Also accept if the student does calculations for the general case and obtains the result

(b) When a current carrying coil is kept in a radial magnetic field the corresponding moving coil galvanometer would have a linear scale 1

Alternatively " In a radial magnetic field two sides of the rectangular coil remain parallel to the magnetic field lines while its other two sides remain perpendicular to the magnetic field lines. This holds for all positions of the coil.”



(a)
(b)